

Concepts of Mathematics

Lectures: 3 & 4

Sections: H I J L M N

Fall 2025

Instructor: Noha Abdelghany(nabdelgh)
OHS: M 10:30–11:30AM & W 10:30–11:30AM
OHS Location: WEAN Hall 8127

Lecture 3: MWF 12–12:50pm in POS A35

Lecture 4: MWF 1–1:50am in POS A35

21–127 SYLLABUS

About the Course

This course introduces the basic concepts, ideas and tools involved in doing mathematics. As such, its main focus is on presenting informal logic, and the methods of mathematical proof.

The subjects above are closely related to the application of mathematics in many areas, particularly computer science.

Topics discussed include a basic introduction to elementary number theory, induction, the algebra of sets, relations, equivalence relations, congruences, partitions, and functions, including injections, surjections, and bijections. A basic introduction to the real numbers, rational and irrational numbers.

Learning objectives

Upon successful completion of this course you should be able to:

1. Write correct, clear and precise mathematical proofs, in an appropriate level of detail, of both familiar and unfamiliar mathematical results in the areas covered;
2. Accurately use standard mathematical notation and terminology in mathematical writing, including logical operators, quantifiers, sets and set operations, functions and relations;
3. Recognize and apply standard proof techniques, including direct proof, proof by contradiction, proof by contraposition, and weak and strong mathematical induction;
4. Accurately recall definitions and state and prove the main theorems in the areas covered;
5. Solve unseen problems in the mathematical areas covered by identifying which proof techniques and results from the course are appropriate and applying them accurately;
6. Typeset basic mathematical documents using $\text{\LaTeX}$, including the mathematical notation used in the course, sections, references and labels, and basic formatting.

Topics and tentative schedule

August

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	25 Intro	26	27 Number Sets	28	29 Logic	30

September

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	Labor day 1 No class	2	Quiz 1 3 Logic	4	5 Logic	6
7	8 Logic	9	Quiz 2 10 Logic	11	12 Sets	13
14	15 Sets	16	Quiz 3 17 Sets	18	19 Sets	20
21	22 Functions	23	Quiz 4 24 Functions	25	26 Functions	27
28	29 Functions	30				

October

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
			Quiz 5 1 Induction	2	3 Induction	4
5	6 Induction	7	Quiz 6 8 Induction	9	10 Catchup	
12	13 Fall Break	14 Fall Break	15 Fall Break	16 Fal Break	17 Fall Break	18
19	20 Relations	21	Quiz 7 22 Relations	23	24 Relations	25
26	27 Relations	28	Quiz 8 29 Cardinality	30	31 Cardinality	

November

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	3 Cardinality	4	Quiz 9 5 Cardinality	6	7 Number Theory	8
9	10 Number Theory	11	Quiz 10 12 Number Theory	13	14 Number Theory	15
16	17 Number Theory	18	Quiz 11 19 Number Theory	20	21 Number Theory	22
23	24 Combinatorics	25	26 Thanksgiving	27	28 Thanksgiving	29

December

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1 Combinatorics	2	3 Combinatorics	4	5 Combinatorics	

Finals

Course materials:

- Our main textbook is
• *An Infinite Descent into Pure Mathematics* by Clive Newstead
ISBN 978-1-950215-00-3

The following books are also relevant and useful:

- *How to Study as a Mathematics Major* by Lara Alcock
ISBN 978-0-19-966131-2
- *Book of Proof (3rd edition)* by Richard Hammack
ISBN 978-0-9894721-2-8
- *Everything You Always Wanted to Know About Mathematics (but Didn't Even Know to Ask)* by Brendan Sullivan
[Download here](#)

Other suggested resources may be announced as the course is in progress—in particular, if you find a useful resource that you think others would benefit from, please contact Prof. Abdelghany.

Assessment and grades

Your performance in the course will be assessed by means of Quizzes and a final exam.

Quizzes

Quizzes are held weekly according to the calendar above. They are 20-minute examinations held in lecture. We will have a total of 11 quizzes, the lowest 3 will be dropped. The quizzes will consist of one or two problems adapted from the previous week's problem sheet. A problem sheet will be posted on a Wednesday and the quiz will be held the following Wednesday. The problems on the quiz will NOT be the exact problems from the problem sheet, but will be similar and will definitely be solvable using the same tools/concepts. The idea is to spend the week learning and understanding how to solve the problems on the problem sheet, which will guarantee you to do well on the quiz.

Problem sheets

Problem sheets are posted as preparation for the quizzes, they are not to be turned in or graded. Even though this "homework" is worth 0% of your grade, it is the part where most of the learning happens. Problem sheets consist of more challenging mathematical exercises based on material introduced in class. It is designed to give you the opportunity to strengthen your understanding, practice using the tools and techniques from the course, and receive feedback. You are meant to struggle in doing these problems, it is likely that you will see some concepts on the problem sheets that haven't been discussed in class, this is essential to develop skills like self-education. You are encouraged to collaborate on these problems with classmates and discuss with your professor and TAs. If you are able to complete all the problems from the problem sheets you will be able to solve the quiz problems with no issue.

Final Exam

Finally, the final exam is a cumulative examination at the end of the course. It is designed to bring together all of the knowledge and skills you have acquired over the course of the semester. We will have a final exam scheduled by the registrar, and you should not leave campus before the final exam.

Course average

Here is the break down of the course overall score:

Component	Weight
Highest 3 quizzes, each	11% · 3
Second Highest 3 quizzes, each	9% · 3
Second lowest 2 quizzes, each	5% · 2
Lowest 3 quizzes (dropped)	0%
Final exam score	30%

The following are the borderlines for letter grades:

A/B: 90% B/C: 80% C/D: 70% D/R: 60%

Flexibility

Sometimes circumstances arise that prevent you from uploading your work on time (or at all). There is flexibility built into the course to deal with such circumstances:

- Enough scores are dropped from the course average computation that you can theoretically miss three weeks of work without a catastrophic impact on your grade. Please note that the dropped scores are in lieu of make-ups and extensions.

Therefore, I will grant NO quiz makeups. If an unforeseen circumstance that is out of your control causes you to miss the final exam, you will be able to take a make-up exam—contact Prof. Abdelghany to discuss next steps.

Academic Integrity

You are responsible of making yourself aware of CMU's Academic Integrity Policies <https://www.cmu.edu/policies/student-and-student-life/academic-integrity.html>. Failing to follow quiz/test instructions and/or attempting to cheat on a quiz/test is an academic integrity violation and could lead to a lower course grade or course failure.

Course resources

Piazza

Piazza is a forum where students can post questions, respond to each others' questions, and get responses from course staff. This is a great way to interact with other students about the course material. If you have a question, you will usually get a response on Piazza faster than if you wait for office hours or a reply to an email.

Office hours

Office hours are your main opportunity to discuss the course material with the course staff. Make the most of them! Even the very best students use office hours to strengthen their knowledge: you do not need to be struggling for office hours to be helpful.

- Have something in mind to discuss or ask, and be as specific as possible.
- Take turns with other students—for example, if you ask a question, allow the other students present to ask a question before you ask a follow-up question.
- Please do not show us or other students your written attempts at homework solutions. It would not be fair for us to provide feedback on written solutions ahead of the submission deadline, but we are happy to discuss your ideas or approach to the problem.
- If in-person office hours are crowded and spilling into the corridor, please respect everyone's space, only stay for as long as you need to, and remember to take your belongings with you when you leave.

University resources

The Student Affairs office maintains a website listing some of the many resources available to you as a CMU student: <https://www.cmu.edu/student-affairs/resources/> – some of these resources are outlined below.

Accommodations

It is very important to me that my methods of instruction and assessment are fair to everyone. If you require any accommodations, including extra time on tests, note-taking services, large-text format materials, alternative test locations, then you should register with the Office of Disability Resources (ODR):

- **Office of Disability Resources**

<https://www.cmu.edu/disability-resources/>

(412) 268-6121 / access@andrew.cmu.edu

Accommodations can only be granted *after* you have registered from ODR and the notification of accommodations has been sent. Other forms of evidence of need for accommodations, such as notes from doctors or emails from parents, cannot be accepted.

Mental health and wellness

If you encounter any difficulties relating to your mental health during the semester, contacting CaPS is a good first step towards getting help if you need it:

- **Counseling and Psychological Services (CaPS)** (confidential resource)

<https://www.cmu.edu/counseling/>

(412) 268-2922

Academic resources

The Student Academic Success Center (SASC) provides a multitude of academic resources for students, including peer tutoring, academic coaching, advising, and guidance for online learning.

- **Student Academic Success Center**

<https://www.cmu.edu/student-success/>

Diversity and inclusion

The Center for Student Diversity and Inclusion offers resources such as mentoring, advising and coaching (and much more), particularly for students in historically underrepresented groups and first-generation students.

More information can be found on the following website:

- **Center for Student Diversity and Inclusion**

<https://www.cmu.edu/student-diversity/>

(412) 268-2150

Discrimination, harassment and sexual misconduct

We share a responsibility to make our community one where everyone feels safe and welcome, both inside and outside the classroom. Discrimination, harassment and abuse (sexual or otherwise) have no place in this course or at CMU, or anywhere else for that matter. Some resources for students regarding discrimination and harassment are as follows.

- **Office of Title IX Initiatives**

<https://www.cmu.edu/title-ix/>

(412) 268-7125 / tix@cmu.edu

- **Hazing Prevention & Response**

<https://www.cmu.edu/hazingprevention/>

- **CMU University Police**

<https://www.cmu.edu/police/>

(412) 268-2323

- **Anonymous reporting**

<https://www.reportit.net/> (username: [tartans](#), password: [plaid](#))

(877) 700-7050